way all of the time—our appetite for risk changes, depending on our recent experiences. For example, imagine we are making a decision to build a new house in California along the San Andreas Fault. If we had just lived through an earthquake, taking on the risk of building a new house on the San Andreas Fault is probably scary, even though the probability of another earthquake hasn't changed. In contrast, when there hasn't been an earthquake in 50 years, building a new house along a fault is not a big deal. In this case, we may perceive the likelihood of an earthquake as being lower than it really is. As this example shows, our perception of risk is not constant and can change based on recent experience. But how does our ever-changing perception of risk matter in the context of financial markets?

Economists often assume that risk aversion, or our willingness to accept risk, is independent of our recent experience. In other words, how we feel about risk will be independent of whether or not we lived through an earthquake 5 minutes ago or 50 years ago. Another assumption economists sometimes make is that risk, often measured in terms of standard deviation, or *volatility*, is relatively constant. These assumptions create a puzzle in the context of extreme stock market drawdowns where the prices of stocks seem to go down further than fundamentals would suggest they should.

What is so puzzling about a 50 percent market drawdown, when, say, the fundamentals suggest the market should only take a 20 percent haircut? Well, as prices move down 20 percent, economists assume that because expected returns have gone up after prices have moved down, and volatility and risk aversion are assumed to be relatively constant, investors should swoop in to buy up cheap shares to ensure they don't drop below 20 percent, which in our example is their so-called fundamental value. But that doesn't happen. If it did, Warren Buffett would be unemployed and markets much calmer than they are. However, stocks can—and have—gone down over 50 percent, and these movements are much more volatile than the underlying dividends and cash flows of the stocks they represent!

How is this possible in a rational market? One approach to understanding this puzzle is by relaxing the assumption that investors maintain a constant aversion to risk. Consider the possibility that investors dynamically change their view on risk after a steep 20 percent drawdown (i.e., they just lived through an earthquake); even though expected returns go up dramatically, risk aversion also shoots up dramatically, which means prices have to go